CTECT 22061

Assignment 1

Day 2 Lab part ii – Expanding LANs

In this lab, you should enhance the topology you previously created during the *Day 2 Lab – Implementing LANs* lab sheet.

Begin by opening your previously saved Packet Tracer file from Day 2 Lab. Using that existing network as your foundation, you should upgrade and extend the topology by incorporating the following additional steps (1, 2 and 3).

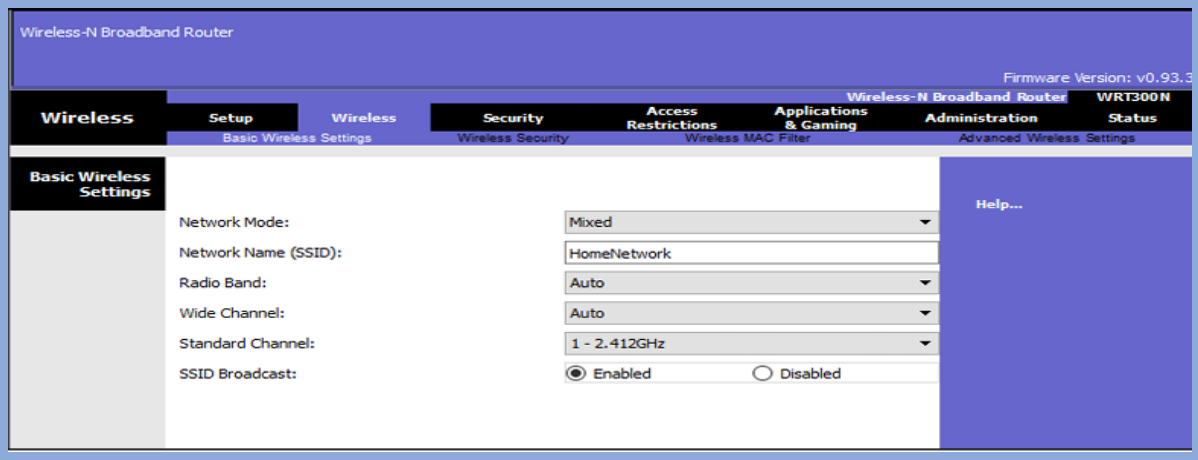
Carefully implement each of the steps below while ensuring proper connectivity and configuration throughout the network.

Step 1: Building a Wireless Network within the LAN

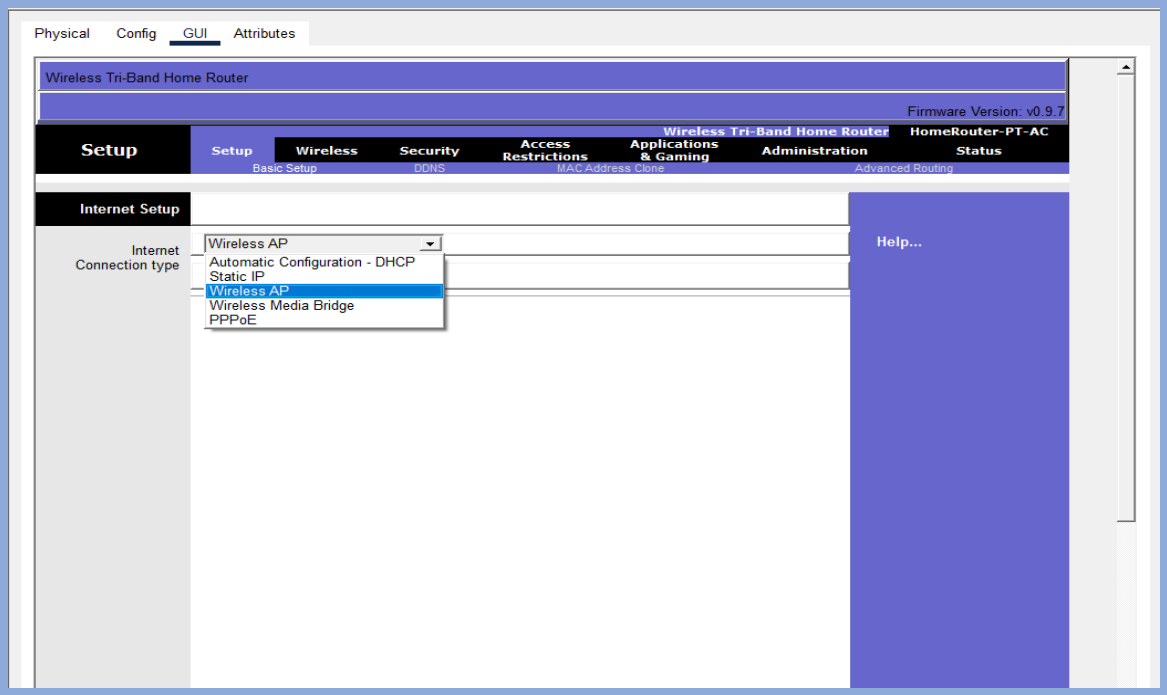
i) Connect a Home Router to the switch using a suitable cable. Click on the home router icon on the Packet Tracer Logical workspace to open the device configuration window.

ii) In the Wireless Router configuration window click on the **GUI tab** to view configuration options for the Wireless Home Router. Next, click on the **Wireless tab** in the GUI to view the wireless settings.

The only setting that needs to be changed from the defaults is the Network Name (SSID). Here, type the name "HomeNetwork_1" as shown in the figure.

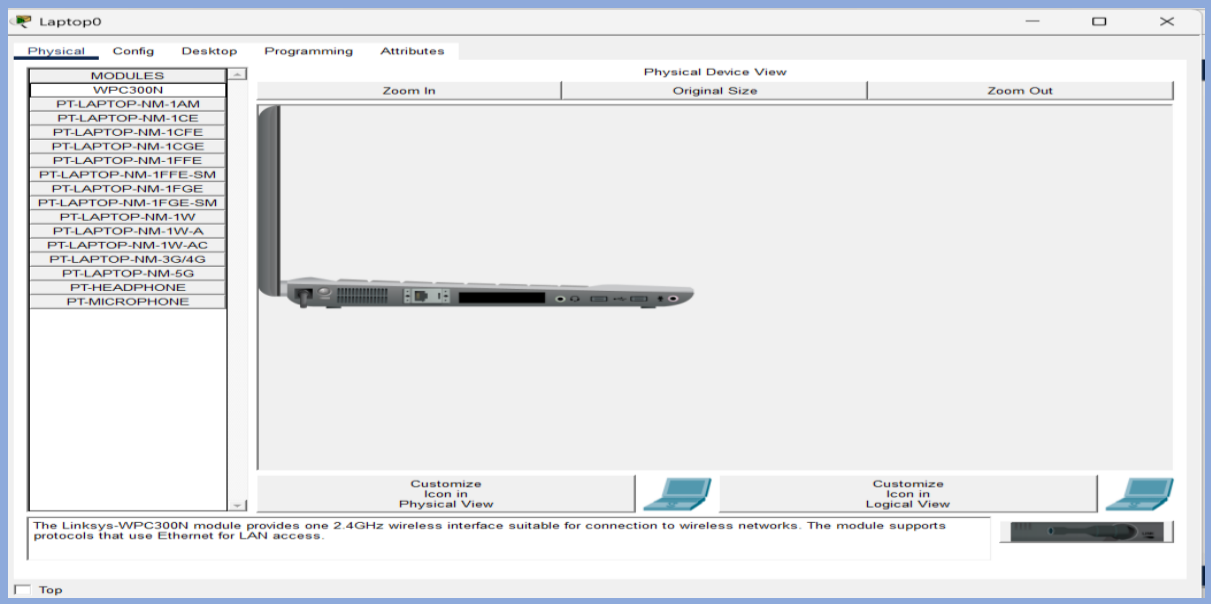


iii) To configure the Internet connection on the Wireless Router, click on the **Setup tab** in the Wireless Router GUI.
Change the internet connection type to 'Wireless AP'.



iv) Take a laptop from the end device list and click on the laptop icon on the Packet Tracer Logical workspace.

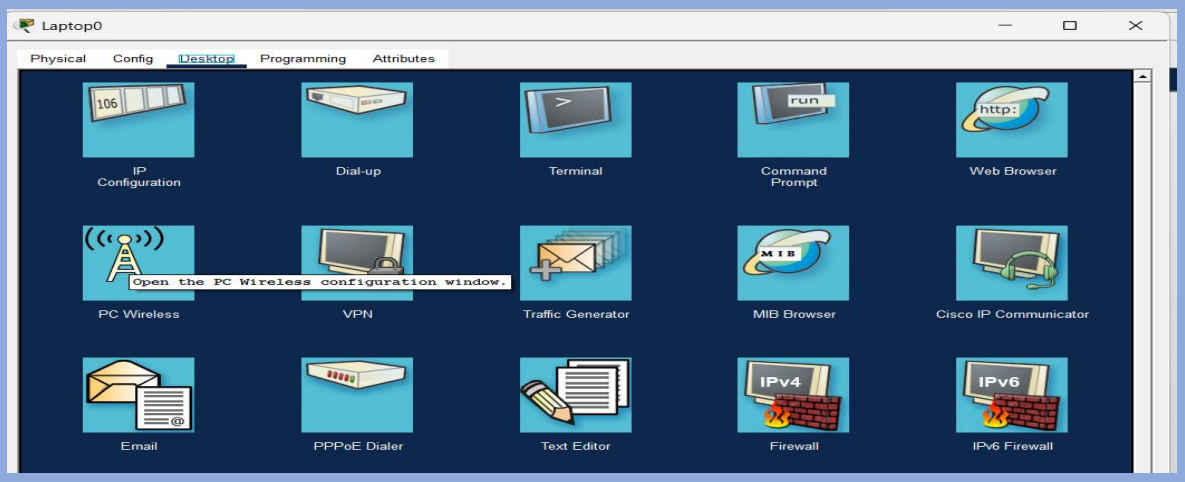
v) In the Laptop configuration windows select the **Physical tab**. In the Physical tab power-off the laptop by clicking the **power button** on the side of the laptop.



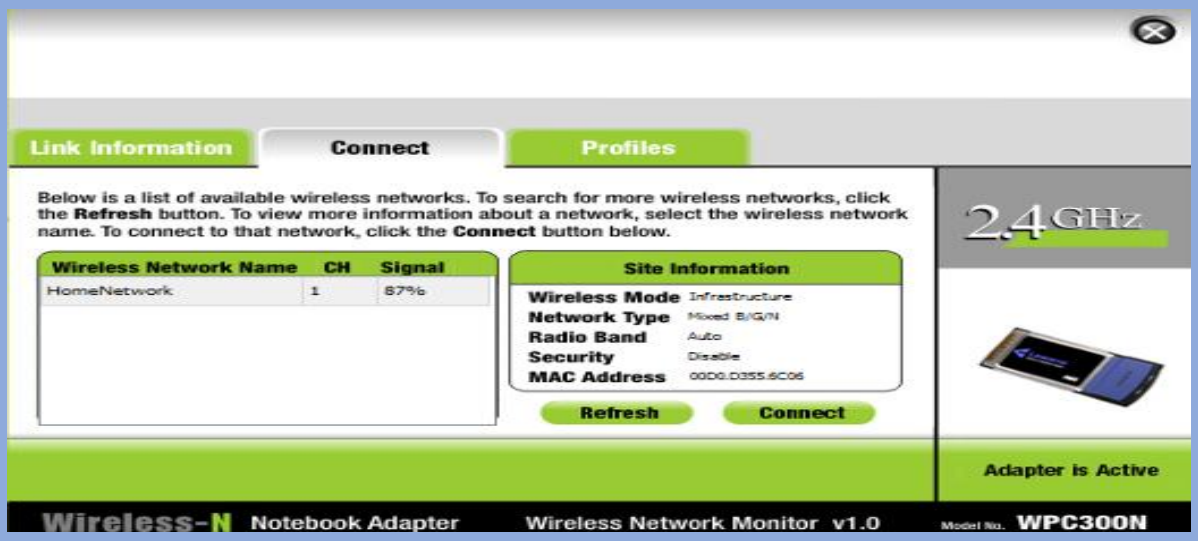
Then remove the currently installed Ethernet copper module by clicking on the module on the side of the laptop and dragging it to the MODULES pane on the left of the laptop window.
Then install the Wireless **WPC300N** module by clicking on it in the MODULES pane and dragging it to the empty module port on the side of the laptop. Power the laptop back on by clicking on the laptop power button again.

With the wireless module installed, the next task is to connect the laptop to the wireless network.

vi) Click on the Desktop tab at the top of the laptop configuration window and select the **PC Wireless** icon.



Once the **Wireless-N Notebook Adapter** settings are visible, select the **Connect** tab. The wireless network "HomeNetwork_1" should be visible in the list of wireless networks as shown in the figure.



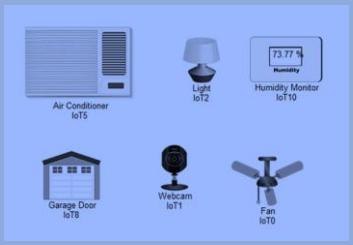
Select the network and click on the **Connect button** found below the Site Information.

vii) Now try to get the IP address of the laptop using ipconfig commands as the PC. Confirm that the IP address belongs to the DHCP pool. And try to pin the laptop with a PC connected to the switch. Confirm whether it is successful.

Step2: Connecting IoT devices with the LAN

i) Take following six IoT devices to the workspace

Humidity monitor
Light
Fan
Webcam
Air conditioner
Garage door



ii) Then install an access point (AP) in the workspace that needs to connect all IoT devices to the network.

iii) Connect the Access Point (AP) to the switch using copper straight through cable.

iv) Click on the AP and go to the **Config tab** and then **port 1** interface. Then name the SSID as “HomeNetwork_2”.

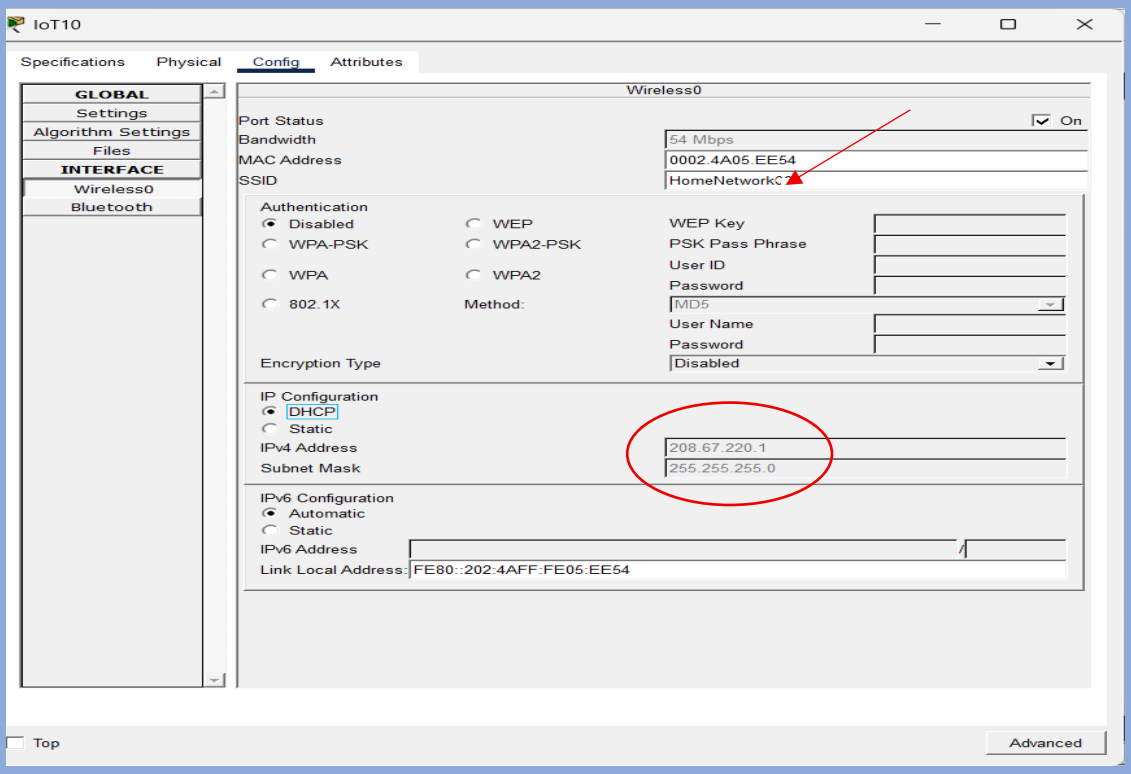
v) Then go to each IoT device separately. You should complete following steps for each IoT device

- Go to the **Config tab** and then **Wireless0** interface.
- Change SSID. (use same SSID name used in AP)

Now you can see the wireless connection between the access point and IoT will be displayed.

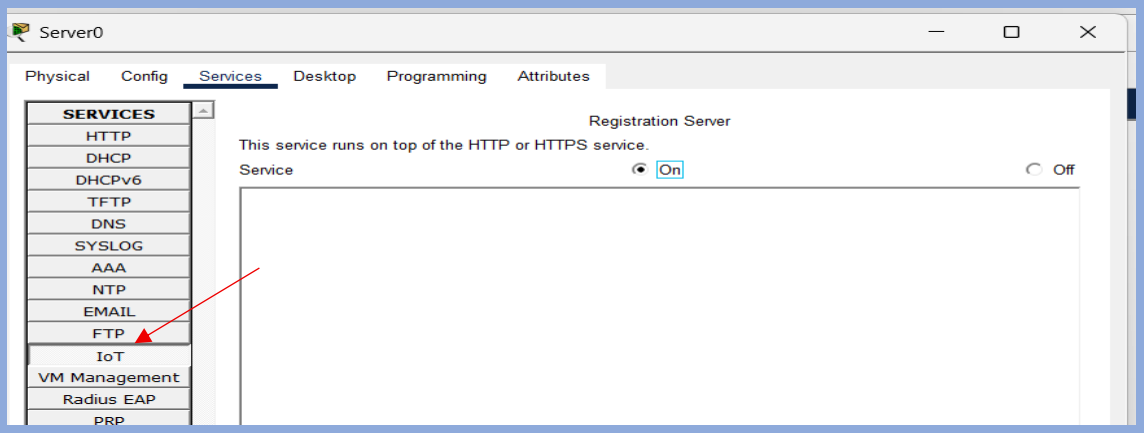
IoT devices' default IPv4 configurations are set to DHCP. You do not need to change them manually. Since the IoT devices have already been connected to the DHCP server through the switch, you can see that all IoT devices have obtained an IP address automatically.

Confirm that the IP addresses belong to the DHCP address range.

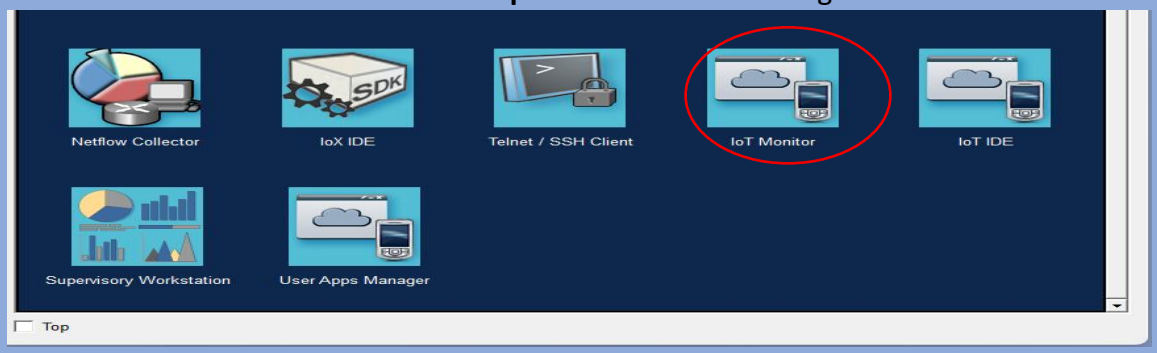


vi) Set up the server as an IoT registration server to manage IoT devices.

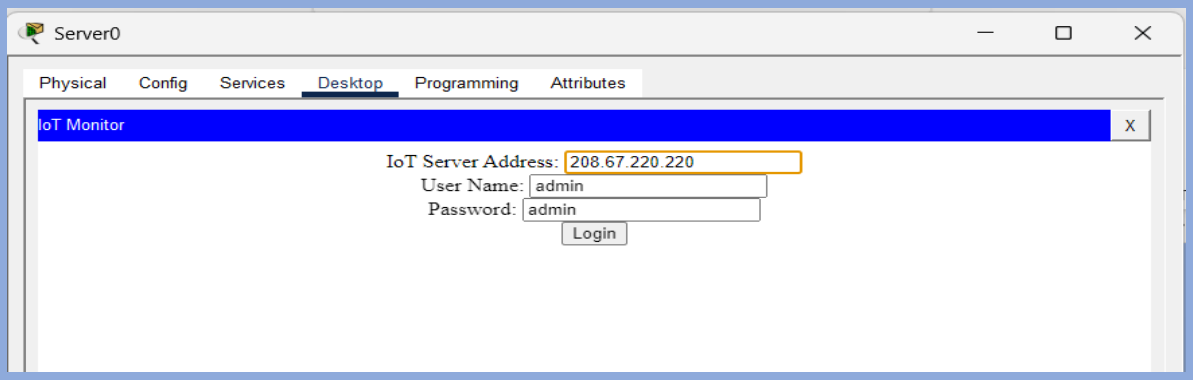
- Go to the **Service tab** on the server. Turn **on** the IoT services.



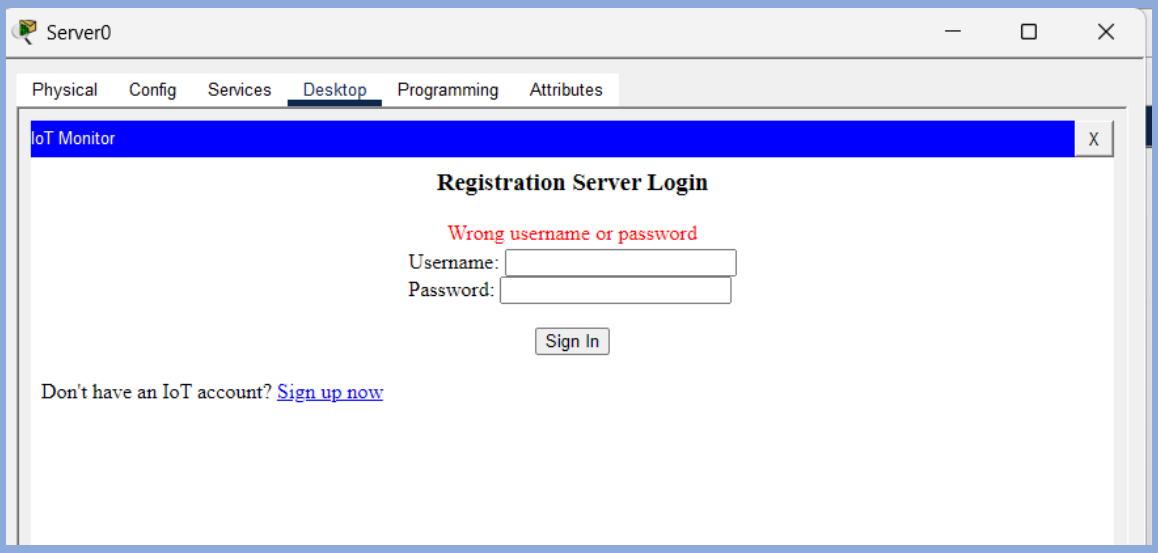
- Then move to the **Desktop tab** in the server. Navigate to 'IoT monitor' section.



- Enter the server IP (208.67.220.220) as the IP and click on the log

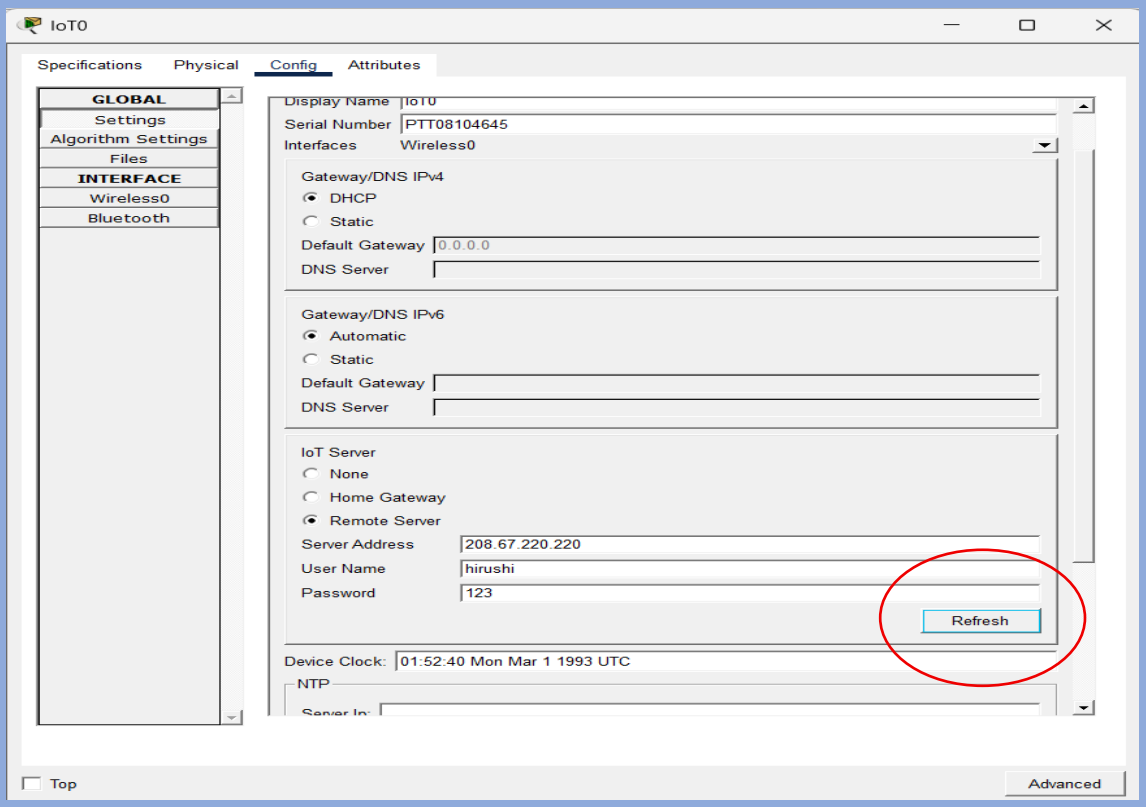


- When you try to log in, a wrong password message will pop up since you do not have an account yet. Then you should go to the 'sign up now' option and create a registration server account. Keep credentials saved.

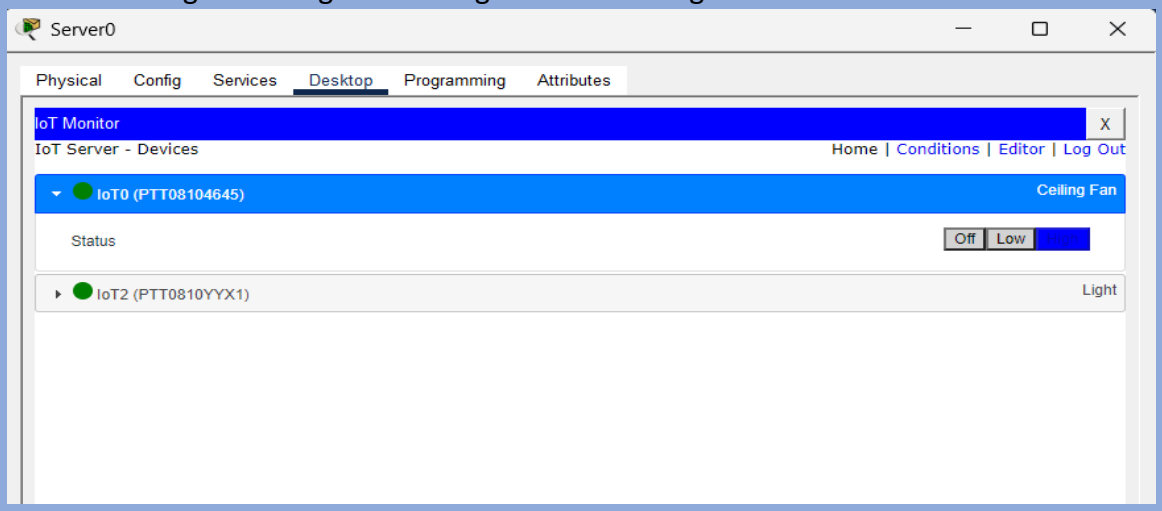


vii) To connect IoT devices to the server, follow the steps below. You should perform the following steps for each IoT device.

- Go to the **Config tab** in the IoT device.
- Go to the **settings.**
- Navigate to the **IoT server** settings section.
- Then select the **remote server** option.
- Enter the server IP (208.67.220.220)
- Enter the credentials for the IoT registration server that you saved.
- Then click on **connect** button. Upon a successful connection, the button will change from 'Connect' to 'Refresh'.



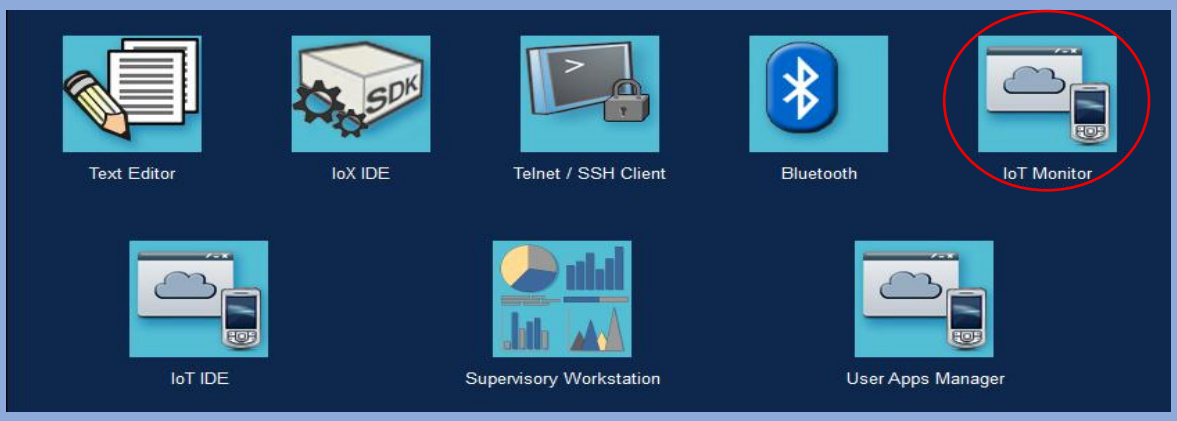
viii) From the server side, you can control all IoT devices after logging in to the registration server again. Change the ceiling fan status as high.



ix) To control IoT devices using smart phones, follow the steps below.

- Add a smart phone to the workspace
- Then connect the smartphone to 'HomeNetwork_2' using the wireless settings, as the other devices were connected above.
- Ensure the smartphone is obtaining an IP address from the DHCP server by enabling DHCP in the IPv4 settings.

- Then in the **Desktop tab** select IoT monitor option. By logging in to the registration server through the smartphone, you can control the IoT devices.



- Perform following changes controlling the IoT devices using mobile phone.
  -Turn on the AC
  -Open the garage door
  -Turn on the web cam
  -Turn on the humidity monitor and measure the value
  -Dim the Light

Step 3: Adding VoIP device to the network

Here, you should perform self-study on how to connect a Voice over IP (VoIP) device to the network and then complete the topology by adding the necessary devices and applying the required configurations.